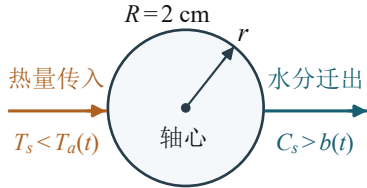
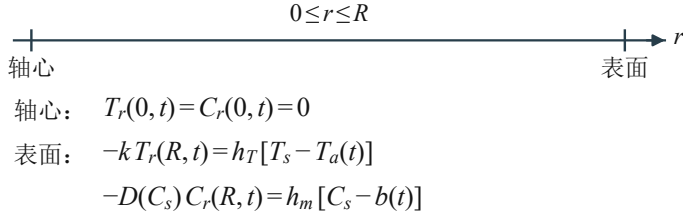


(a) 中截面与通量方向



箭头示意本问预热工况

(b) 径向区间与边界条件



$$T_s = T(R, t), \quad C_s = C(R, t)$$

$b(t)$: 有效边界值